

# Zach Zukowski

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## AI GOVERNANCE · CONTROL-LAYER ACCOUNTABILITY · DILIGENCE FRAMEWORKS

- 22 MCP + API connections · Diligence-doc review 4hr to 1hr
- 8 SSRN papers · 14 federal comment letters · FSB AI consultation response
- 16 USPTO patent applications across 8 families (Canonical-State Architecture for multi-agent AI research)

### SUMMARY

AI-augmented research and governance-infrastructure builder who turns control-layer and operator-accountability research into machine-readable diligence systems, now extending that accountability lens from stablecoin gateways to AI deployment. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: research-infrastructure engineering, control-layer accountability, machine-readable diligence, agentic workflow automation · MCP server design, Monte Carlo, difference-in-differences, coalition-power analysis · Python, R, DuneSQL.

### EXPERIENCE

#### Founder · Tokenization Systems

*Feb 2026 to Present*

- Authored an eight-paper SSRN-live research program covering the [Control Layer Intensity Index](#) (CLII) for gateway concentration, the [Minimum Viable Equivalence Pack](#) (MVEP) for institutional tokenization diligence, programmable monetary-policy design ([Adaptive Tokenomics](#)), [governance concentration](#), and [operational risk in token economies](#).
- Contributed fourteen comment letters across a federal regulatory program (OCC bank stablecoin engagement [RIN 1557-AF41](#) plus [supplemental control-layer extension](#); [Treasury GENIUS Act state similarity](#); FDIC payment-stablecoin reserve standards [RIN 3064-AG19](#) and stablecoin-subsidiary issuance approvals [RIN 3064-AG20](#); FinCEN AML/CFT [FINCEN-2026-0034](#); FinCEN-OFAC PPSI sanctions [FINCEN-2026-0100](#); joint banking AML/CFT [91 FR 18304](#)); first international standard-setter filing: [FSB AI Sound Practices consultation response](#) (Jul 2026), the first external deployment of the MVEP-AI / CLII-AI operator-accountability stack; [Routing the Dollar](#) submitted to the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); [Governance Concentration](#) accepted for publication at Frontiers in Blockchain.
- Working dialogue with capital-markets infrastructure operators on stablecoin, deposit-token, tokenized-deposit, and tokenized-equity deployment.

#### Senior Investment Analyst, Digital Assets · Borderless Capital (RIA; first hire)

*Dec 2019 to Feb 2026*

##### Research Function & Automation

- Built an automated SEC-filing and press-release monitoring pipeline (n8n + Apify) covering 30+ Digital Asset Treasury Stocks (DATs), tracking coins held, cash, debt, and share counts (float, adjusted, fully diluted) per issuer to compute multiple on NAV (mNAV) apples to apples across the sector; analyzed ~75 tokenization offerings in 2025 and invested in 18; AI-assisted diligence workflows reducing subscription-document review from ~4 hours to ~1 hour per deal.
- Built the firm's research function from scratch; led a four-analyst team and personally produced ~25 of its ~125 long-form publications annually; taught 100 teams across nine accelerator programs (2021 to 2025).

##### Governance Standards Advisory

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.
- Advised 100 portfolio companies 1-on-1 on token mechanism design (stablecoin reserves, governance-token allocation and incentive design, yield wrappers, burn/emission dynamics); standards defined in [Introduction to Tokenomics](#).
- Built quantitative token-economic models forecasting emissions, fee capture, pricing and subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges; supply-equilibrium and slashing dynamics supporting Layer 1 business-model and mechanism-design advisory.

##### Investment & Conviction Calls

- Sourced the **LayerZero seed round** investment for Borderless, returning ~80x; the underlying cross-chain interoperability thesis preceded the firm's Cross-Chain Ecosystem Fund by two years.
- Predicted Thorchain's **\$200M insolvency 13 months in advance**; Borderless held no exposure when the predicted failure mode materialized.
- Shipped 250 portfolio investments (~25% deployment rate) across ~10 institutional funds (~\$500M AUM) from a 1,000+ deal pre-seed-to-Series-C funnel, with build/partner/hold recommendations to committee.

## SELECTED RESEARCH

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- **Canonical-State Architecture.** Filed sixteen US provisional patent applications in 2026: a five-element coordination system for concurrent AI-agent sessions on a shared version-controlled repository, integrating per-session staging-index isolation, a multi-mode verification engine, a commit-time canonical-state coherence gate, recursive self-validation of discipline codifications, and an operator-confirmed promotion gate; validated across ~9,600 commits and 16 or more parallel AI coding sessions over ~4 months (as of 2026-07-15), with a companion 73-page methodology preprint. The applications extend the architecture to role-routed cross-session coordination, actual-touched-file collision detection, and a self-describing verification registry, plus separate applications for evidence-grounding of AI-generated assessments and cited values.
- **[Auditing Governance Concentration Beyond Token Allocation.](#)** Accepted for publication at Frontiers in Blockchain. Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's  $d = 0.65$ , a directionally robust medium effect).
- **[Operational Risk in Token Economies.](#)** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 31 program objectives) mapped to a 624-scenario battery; Lido V3 staking-vault case study.
- **[Adaptive Tokenomics.](#)** Industrial control engineering (PID controller) applied to token-supply emission as programmable monetary-policy mechanism; 806-run agent-based simulation battery; cuts worst-case node-count variance ~6x under competitor-shock attack.
- **[Who Burns the Tokens?](#)** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.
- **[Routing the Dollar.](#)** Submitted to Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways including Visa Direct-adjacent paths.
- **[Minimum Viable Equivalence Pack.](#)** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at \$40B scale. Editor-invited submission to the Journal of Financial Regulation.
- **[The Control Layer of Agentic Payments.](#)** First control-layer measurement of live agentic payments (x402): facilitator concentration above the antitrust highly-concentrated threshold on every chain measured, subsidy and proxy pipelines exposed by buyer cardinality, and a composition chain with no accountable layer; anchors the FSB AI-consultation response.
- **[FSB AI Sound Practices response.](#)** Filed 2026-07-01; the program's first international standard-setter filing. Three practice additions (composition-chain documentation, execution-authority evidence, dependency-concentration monitoring) plus the invited nonbank case study on settlement facilitators.
- **[Seven Dollars.](#)** Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).

## TECHNICAL

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**Methods:** policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

**Data:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

**Stack:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino), JavaScript / TypeScript; 22 MCP and API connections wrapping the data sources above plus GitHub; Claude.ai + Claude Code + Cursor with 36-skill library and 17-file living-knowledge architecture; n8n, Apify, Git / GitHub

## RECOGNITION

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- AI-governance focus group at Foresight Institute Vision Weekend
- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- **[Introduction to Tokenomics](#)** article publicly endorsed by Binance founder CZ on day of publication

## EDUCATION

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**High Point University** | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President  
Lingnan University, Hong Kong (Exchange)

2018